Discovery Lab

Exam 3 Revision

Lower Secondary Stage 2 · Science · Lessons 12 to 16

Reproduction, pregnancy, the menstrual cycle, acids and hazards, and the pH test.



Lesson 12: Reproduction

Lesson 13: Pregnancy

Lesson 14: The Menstrual Cycle

Lesson 15: Acids and Hazards

Lesson 16: The pH Test

Name: _____ Class: _____ Date: _____

LESSON 12

12 Reproduction

How living things make new life, and why it matters for endangered species.

Key ideas

Sexual reproduction needs **two parents**.

A **sperm** (male) and an **egg** (female) join in **fertilisation**.

Offspring are **genetically different** from their parents: this is **variation**.

Reproduction passes on genes and keeps the **species** alive.

Endangered species are at risk of extinction. Helpers use **captive breeding**, **artificial insemination**, **protected nests** and **relocation**.

Oviparous = lay eggs, **viviparous** = live young, **ovoviviparous** = eggs hatch inside the mother.

Species	Reproduction challenge
Sea turtle	Eggs are often eaten or destroyed by predators
Giant panda	Females are fertile for only a few days a year
Orangutan	Needs safe forests to raise babies
Rhino	Very long pregnancy (about 16 months)
Cheetah	Poor genetic diversity in the wild

DID YOU KNOW?

A giraffe's pregnancy lasts about 15 months, longer than a school year!

Check yourself

- [ ] What is fertilisation?
- [ ] Why do brothers and sisters not look identical?
- [ ] What happens if an egg is not fertilised?
- [ ] Why is variation useful in nature?
- [ ] How does sexual reproduction help endangered species?

REVISION TASK

Match the terms sperm, egg, fertilisation, variation and sexual reproduction with their meanings, then sketch a sperm cell and an egg cell.

LESSON 13

13 Pregnancy

From a fertilised egg to a baby in about nine months.

Key ideas

Fertilisation happens in the **oviduct**. The fertilised egg travels to the **uterus** and attaches.

Zygote (one cell) → **embryo** (ball of cells) → **foetus** (organs, bones, heartbeat).

The **placenta** brings oxygen and nutrients and removes waste.

The **umbilical cord** links the baby to the placenta.

The **amniotic sac** is full of fluid and protects the baby.

A human pregnancy lasts about **9 months** (about **280 days** or **40 weeks**).

Month	What happens?
1	The fertilised egg implants in the uterus
3	The baby has eyes, arms, legs and a heartbeat
6	The baby can move and hear
9	The baby is fully developed and ready to be born

DID YOU KNOW?

If the placenta stopped working, the baby could not get enough oxygen or nutrients and waste would build up.

Check yourself

- [ ] How does pregnancy begin?
- [ ] What does the placenta do?
- [ ] What is the difference between an embryo and a foetus?
- [ ] What protects the baby inside the uterus?
- [ ] What changes happen by month 3 and month 9?

REVISION TASK

Draw a timeline from month 1 to month 9 and add one fact for each stage.

LESSON 14

14 The Menstrual Cycle

How the body prepares for a possible pregnancy every month.

Key ideas

The cycle lasts about **28 days** (it can vary).

A **period** (menstruation) lasts about **3 to 7 days**. It is normal and healthy.

Most girls get their first period between **9 and 13**.

Ovulation: an egg is released from the ovary around **day 14**.

If the egg is not fertilised, the uterus lining breaks down.

Four key hormones: **FSH**, **LH**, **oestrogen**, **progesterone**.

Days	What happens?
1–5	Menstruation: the lining breaks down and is lost
6–13	A new lining builds and an egg starts to mature
14	Ovulation: an egg is released
15–28	The lining stays thick, waiting for a fertilised egg



DID YOU KNOW?

A person who menstruates has about 450 periods in a lifetime: over 10 years!

Check yourself

- [ ] What is a period?
- [ ] What happens on day 14?
- [ ] Why does the lining break down?
- [ ] Which hormone causes ovulation?
- [ ] How are the cycle and pregnancy connected?

REVISION TASK

Match each hormone to its job: FSH matures eggs, LH triggers ovulation, oestrogen builds the lining, progesterone keeps it thick.

LESSON 15

15 Acids and Hazards

Acids, alkalis and the symbols that keep us safe.

Key ideas

Acids: sour, pH **below 7** (vinegar, lemon juice; strong: hydrochloric acid).

Alkalis: soapy, pH **above 7** (soap, toothpaste; strong: bleach, sodium hydroxide).

Strong acids and alkalis can be **corrosive**.

Corrosive: damages skin or eyes. **Flame**: flammable. **Exploding bomb**: explosive.

Exclamation mark: harmful or irritant. **Biohazard**: dangerous bacteria, viruses or toxins.

Hazard symbols warn people and help prevent accidents.

Substance	Type	Hazard
Bleach	Alkali	Corrosive to skin
Vinegar	Weak acid	Safe, but sour
Battery acid	Strong acid	Dangerous: corrosive
Washing-up liquid	Weak alkali	Mild irritant
Oven cleaner	Strong alkali	Can cause burns

DID YOU KNOW?

The strongest acid made in a lab, fluoroantimonic acid, is billions of times stronger than stomach acid and can dissolve glass.

Check yourself

- [ ] What is the difference between acids and alkalis?
- [ ] What does the corrosive symbol mean?
- [ ] Name a real corrosive substance.
- [ ] Why are hazard symbols important?
- [ ] Which household product is a weak alkali?

REVISION TASK

Draw five hazard symbols, write what each means and give one example of each.

LESSON 16

16 The pH Test

Measuring how acidic or alkaline a substance is.

Key ideas

The pH scale goes from **0** (strong acid) to **14** (strong alkali). **pH 7 is neutral**, for example pure water.

Universal indicator: red = strong acid, green = neutral, purple = strong alkali.

Litmus paper: red in acid, blue in alkali, no change in neutral.

Red cabbage is a natural pH indicator.

Typical pH: lemon juice 2, vinegar 3, pure water 7, soap 9, bleach 12.

A fair, safe test: same amount of liquid, same indicator, safety glasses, record in a table.

Substance	pH	Acid, alkali or neutral?
Lemon juice	2	Acid
Vinegar	3	Acid
Pure water	7	Neutral
Soap solution	9	Alkali
Bleach	12	Alkali



DID YOU KNOW?

Red cabbage water changes colour in acids and alkalis. You can make your own pH indicator at home!

Check yourself

- [ ] What is the pH of a neutral substance?
- [ ] What colour is universal indicator at pH 1?
- [ ] Which household item has a pH of about 12?
- [ ] How can you test if a liquid is acidic or alkaline?
- [ ] Which natural item can be used to test pH?

REVISION TASK

Draw the pH scale from 0 to 14, colour it in and add one substance for the acid, neutral and alkali sections.

FINAL PRACTICE

Final practice

Tick each box when you can answer without looking.

Final practice

- [ ] Why is variation useful in nature?
- [ ] What connects the baby to the mother inside the uterus?
- [ ] Describe what happens during ovulation.
- [ ] Why is vinegar an acid?
- [ ] How do you test the pH of a liquid?

CHALLENGE

Write a short paragraph explaining how the menstrual cycle and pregnancy are connected.

Revision checklist

Revision task	Done?
Reviewed the Canvas slides for Lessons 12–16	☐
Answered every "Check yourself" question	☐
Created diagrams, timelines and tables	☐
Practised the long answers on the website	☐
Played the 3D labs and the pH Runner	☐
Asked for help on anything confusing	☐

Try it live · Practise with 3D labs, games and a practice exam at **morehumaned.com/discovery-lab**

ANSWER KEY

Answer key

Check your own answers against these. If one is hard, go back to the lesson.

Lesson 12: Reproduction

1. A sperm cell joins an egg cell to make a fertilised egg.
2. They inherit a different mix of genes from both parents.
3. No embryo forms and the egg breaks down.
4. Some individuals may survive if the environment changes.
5. It makes variation, and helping animals breed keeps the species from dying out.

Lesson 13: Pregnancy

1. A sperm fertilises an egg in the oviduct and the fertilised egg attaches to the uterus lining.
2. It brings oxygen and nutrients and removes waste.
3. A foetus has organs, bones and a heartbeat.
4. The fluid-filled amniotic sac.
5. Month 3: eyes, arms, legs, heartbeat. Month 9: fully developed, ready to be born.

Lesson 14: The Menstrual Cycle

1. The loss of the uterus lining and a little blood when there is no fertilised egg.
2. Ovulation: an egg is released from the ovary.
3. The body no longer needs it because the egg was not fertilised.
4. LH.
5. The cycle prepares the uterus so a fertilised egg can attach and grow.

Lesson 15: Acids and Hazards

1. Acids have pH below 7 and taste sour. Alkalis have pH above 7 and feel soapy.
2. It damages skin or eyes.
3. Battery acid, bleach or oven cleaner.
4. They warn people of dangers and help prevent accidents.
5. Soap or washing-up liquid.

Lesson 16: The pH Test

1. pH 7.
2. Red.
3. Bleach.
4. Use an indicator such as litmus paper or universal indicator.
5. Red cabbage.

Final practice

1. Some individuals may survive if the environment changes.
2. The placenta (and the umbilical cord).
3. An egg is released from the ovary, around day 14.
4. Its pH is below 7.
5. Add an indicator such as universal indicator and compare the colour with the chart.

Discovery Lab

You are going to do great! Use this booklet to plan your study time, then bring it all together.

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